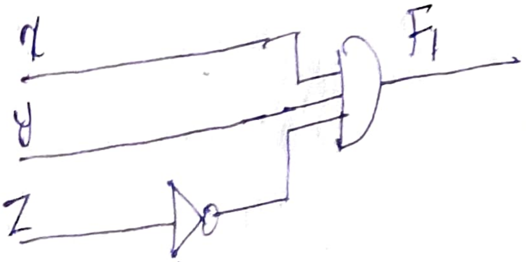
• Any Boolean function can be represented in a truth table. The number of rows in the table is 2^n, where n is the number of binary variables in the function. The 1's and 0's combinations for each row is easily obtained from the binary numbers by counting from 0 to 2^n-1. For each row of the table, there is a value for the function equal to either 1 or 0.

• Q) Is an algebraic expression of a given Boolean function unique? In other words, Is it possible to find two algebraic expressions that specify the same function?

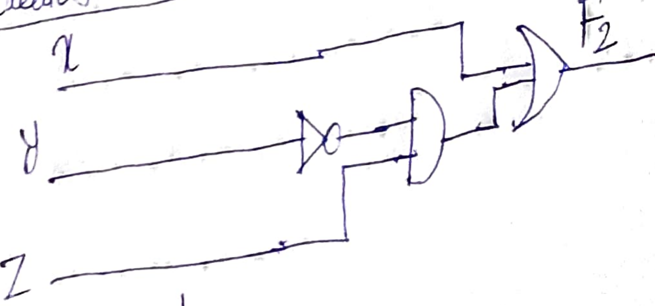
↳ YES!!

• From table 2-2, we find that F_4 is the same as F_3 since both have identical 1's and 0's for each combination of values of the three binary variables. In general, two functions of n binary variables are said to be equal if they have the same value for all possible 2^n combinations of the n variables.

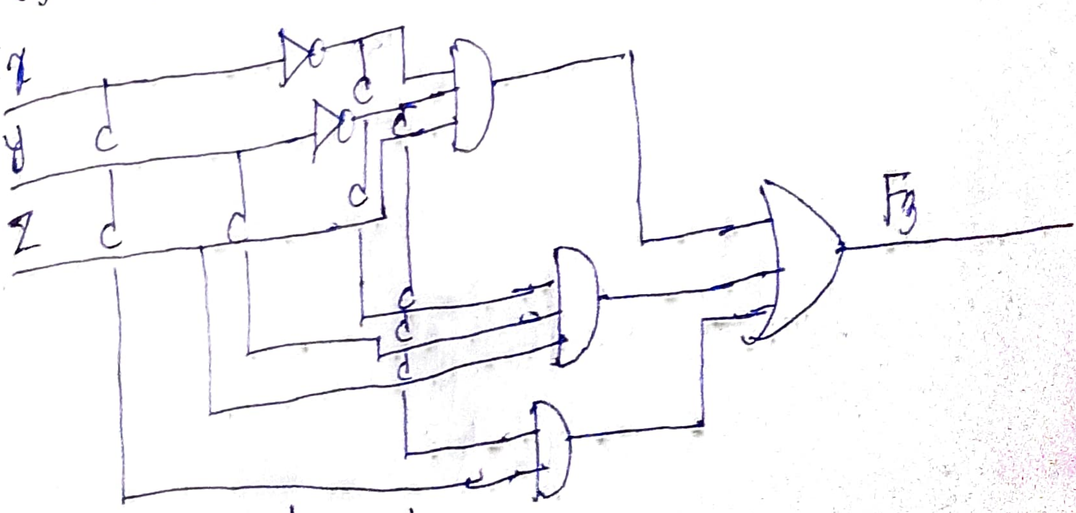
Figure 2.4 Implementation of Boolean function with gates



(a) $F_1 = xyz'$



(b) $F_2 = x + y'z$



(c) $F_3 = x'y'z + x'yz + xy'$